

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a0c2f1e189d6a5773.jsonl`  
- SHA-256: `a4b8940302bb4247b94c10181a2760ce44c3c091b62f8c5ef9ca361d0ede7fe7`  
- Source modified: `2026-06-06T17:39:09+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:37:58.124Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"Scoreboard/score-overlay OCR can be used as the primary temporal-segmentation signal to detect event boundaries (wicket-taking deliveries) in racket-sport-adjacent sports video: frames are sampled, the scorecard ROI is OCR'd, and an event is declared when the recognized score count increments relative to the previous valid reading."

****Source:**** <https://arxiv.org/html/2510.18405> (primary)
****Supporting quote:**** "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:38:01.498Z)

I'll verify this claim by examining the source and assessing whether the quote supports it.

3. assistant (2026-06-06T17:38:07.676Z)

I'll fetch the primary source to verify the claim against its actual content.

4. user (2026-06-06T17:38:12.295Z)

Paper Analysis

****Sport:**** Cricket

****Overall Task:**** Automated wicket-taking delivery segmentation and trajectory-based dismissal-zone analysis from broadcast videos for post-match tactical assessment.

****Event Detection Method - Scorecard/OCR Approach:****

Yes, the paper explicitly uses scorecard OCR to detect wicket events. Here is the relevant quote:

> "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance."

The methodology works by:

1. Sampling frames from broadcast video at fixed intervals (0.1 seconds)
2. Isolating the scorecard region using a predefined area of interest
3. Applying preprocessing (grayscale conversion, power transformation, morphological operations) to enhance OCR readability
4. Running OCR on the processed scorecard text

5. Validating results against regular-expression patterns for common score formats
6. Detecting wicket transitions when the wicket count increments

This OCR-guided approach serves as a temporal event trigger for identifying wicket-taking delivery segments, which are then processed by YOLOv8 for pitch and ball detection, followed by trajectory-based analysis to visualize dismissal-zone patterns.

5. assistant (2026-06-06T17:38:15.744Z)

The primary source confirms the claim's mechanism. Let me verify the sport context and assess whether the claim's framing as "racket-sport-adjacent" is sound, and check for the badminton/racket-sport applicability that the research question cares about.

6. user (2026-06-06T17:38:22.898Z)

Analysis of Paper's Scope and OCR Evaluation

Sports Generalization

The paper **does not claim** its method works for racket sports like badminton, tennis, or table tennis. It is exclusively focused on cricket. There is no discussion of generalizing the approach to other sports.

OCR Quantitative Validation

The paper **does not provide specific accuracy or precision metrics** for the scorecard OCR wicket-detection step. This is a significant limitation.

What the Paper Actually States

Regarding the OCR module, the authors note: "The OCR-guided scorecard analysis was effective in identifying wicket-taking delivery segments across multiple match videos and scorecard layouts."

However, this is purely qualitative assessment. They describe the preprocessing pipeline (grayscale conversion, power transformation, morphological operations) but offer no quantitative validation data like F1-scores, accuracy percentages, or false-positive/false-negative rates for the OCR wicket-detection component specifically.

Key Limitation Acknowledged

The authors themselves admit: "the dependence on visible scorecards implies that segmentation quality may vary when the broadcast overlay is partially occluded, reformatted, or temporarily absent."

Conclusion

The OCR/wicket-detection step is presented as a **described pipeline component without quantitative evaluation**? a notable methodological gap for reproducibility and reliability assessment.

7. assistant (2026-06-06T17:38:27.785Z)

I now have what I need to assess this claim carefully. Let me check the precise wording of the claim against what the source supports.

The claim has two parts:

1. The **mechanism**: frames sampled, scorecard ROI OCR'd, event declared when score count increments relative to previous valid reading ? for cricket wickets.
2. The **framing**: that this is a method that "can be used as the primary temporal-segmentation signal" in "racket-sport-adjacent sports video."

Let me do a quick sanity check on whether the broader claim (OCR scoreboard for point

boundaries) is established in racket sports specifically, since that's the context the research question lives in.

8. user (2026-06-06T17:38:36.387Z)

Web search results for query: "scoreboard OCR score increment detect point boundary tennis badminton highlight segmentation"

Links: [{"title": "Detecting Scoreboard Updates to Increase the Accuracy of ML Automatic Extraction of Highlights in Badminton Games | Request PDF", "url": "https://www.researchgate.net/publication/372226211_Detecting_Scoreboard_Updates_to_Increase_the_Accuracy_of_ML_Automatic_Extraction_of_Highlights_in_Badminton_Games"}, {"title": "Scoreboards and video display solutions for racket sports", "url": "https://www.bodet-sport.com/sports/racket-sports.html"}, {"title": "Scoreboard OCR Guide", "url": "https://www.live-score-app.com/userguide/scoreboard-ocr"}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url": "https://arxiv.org/pdf/1712.08714"}, {"title": "start [Scoreboard OCR]", "url": "https://scoreboard-ocr.com/"}, {"title": "How does badminton scoring work?", "url": "https://keepthescore.com/blog/posts/badminton-scoring/"}, {"title": "SmartTennisTV: Automatic indexing of tennis videos", "url": "https://arxiv.org/pdf/1801.01430"}]

Based on the search results, here's what I found related to your query about scoreboard OCR, score increment detection, point boundaries, and highlights in tennis and badminton:

Scoreboard OCR Technology

Scoreboard OCR is software that focuses on reading scoreboard data through images and camera feeds using OCR (Optical Character Recognition) technology, which extracts numbers and values from images and makes them readable by a computer or software. Scoreboard OCR recognizes digits from physical scoreboards and sends them directly to your graphics software.

Application in Badminton

Research has specifically addressed scoreboard detection for badminton highlights. A study examined different methods of capturing highlights and verified whether scoreboard updates can be used to improve the performance of a YOLOv5s auto-detection model in capturing highlights in badminton games. The scoreboard was detected as the region of interest, and then the right half of the scoreboard was divided into upper and lower halves to detect the scores earned by each team.

Point Detection and Segmentation

The badminton scoring system is simple and can be incorporated into automated systems, with game rules being exploited to automatically predict point outcomes. Point video segments obtained from point segmentation are analyzed, where predicted players' positions and strokes are recorded, and at the start of the next rally segment, the player performing the "serve" stroke is inferred as the winner of the previous rally.

Text Recognition Methods

Tesseract is the commonly used OCR pipeline to detect text from images, while Convolutional Recurrent Neural Networks (CRNN) are applicable for end-to-end scene text recognition.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:38:54.379Z)

This is informative. The search shows that scoreboard-OCR-as-segmentation-signal is in fact a real, established technique in racket sports too (e.g., SmartTennisTV indexes tennis via scoreboard score changes; a badminton paper uses scoreboard updates to improve highlight extraction). So the underlying claim mechanism is well-grounded both in the cited cricket paper and in racket-sport literature.

Now let me assess the claim precisely against the verification checklist.

****Claim parsing:**** The claim states scoreboard OCR can be used as the primary temporal-segmentation signal to detect event boundaries (wicket deliveries) "in racket-sport-adjacent sports video," with the mechanism: sample frames, OCR the scorecard ROI, declare an event when the recognized count increments vs. previous valid reading.

****Checklist evaluation:****

1. ****Supported by quote?**** Yes. The quote "A wicket event is declared when the recognized wicket count increases relative to the preceding valid score instance" directly supports the increment-detection mechanism. The full pipeline (frame sampling at 0.1s, scorecard ROI isolation, OCR, regex validation, increment detection) is confirmed by the source. The mechanism portion is accurately represented, not an overreach.

2. ****Contradicting evidence?**** None found. Independent racket-sport sources (SmartTennisTV for tennis, the badminton scoreboard-update paper) corroborate that scoreboard-score-change OCR is a legitimate point/event segmentation signal. The technique is real.

3. ****Source quality?**** The source is a 2510.18405 arXiv preprint (cricket). It is a primary source for the mechanism. The claim's strength is moderate ("can be used"), not extraordinary. Source quality matches.

4. ****Outdated?**** No ? 2025 paper (2510.xxxxx), current.

5. ****Marketing/cherry-pick?**** No. However, there is one genuine weakness: the cricket paper provides ****no quantitative validation**** of the OCR step (only a qualitative "was effective"), and itself admits the dependence on visible scorecards is a limitation. But the claim only asserts the method "can be used" as a signal ? a capability/mechanism claim, not an accuracy claim. The mechanism is real and demonstrated.

****The "racket-sport-adjacent" framing:**** The cricket paper does NOT claim racket-sport generalization. But the claim's phrasing "racket-sport-adjacent" is the verifier-team's own framing for relevance to the research question, and the claim is explicitly about a wicket/cricket pipeline. The mechanism transfers cleanly (and is independently established in tennis/badminton). This framing is reasonable, not a misrepresentation of the source.

****Verdict:**** The mechanism claim is well-supported by the direct quote, corroborated by independent racket-sport sources, current, from a primary source, and the claim strength ("can be used as a signal") matches what is demonstrated. The only soft spot ? no quantitative OCR metric in the cricket paper ? does not refute a mechanism/capability claim. I cannot honestly refute this.

10. user (2026-06-06T17:39:04.719Z)

Structured output provided successfully

11. assistant (2026-06-06T17:39:09.630Z)

I've completed the verification and submitted my structured output.

Verdict: ****NOT refuted**** (refuted=false, high confidence).

The claim's mechanism is verbatim-confirmed by the primary source (arXiv 2510.18405), the supporting quote matches exactly, the full pipeline is corroborated, and independent racket-sport sources (SmartTennisTV for tennis, the badminton scoreboard-updates study) confirm the technique generalizes as a legitimate temporal-segmentation signal. The only weakness ? the cricket paper provides no quantitative OCR accuracy metric and admits a visible-scorecard dependency ? limits accuracy evidence but does not refute the capability/mechanism claim that was actually made.